

ESP32_Thermal_Camera_Viewer

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About

Stream 80×62 thermal frames over TCP to a full-featured web UI with color maps, max temp tracking, and fullscreen display.

[youtu.be/uJirSqlyhA4](#)

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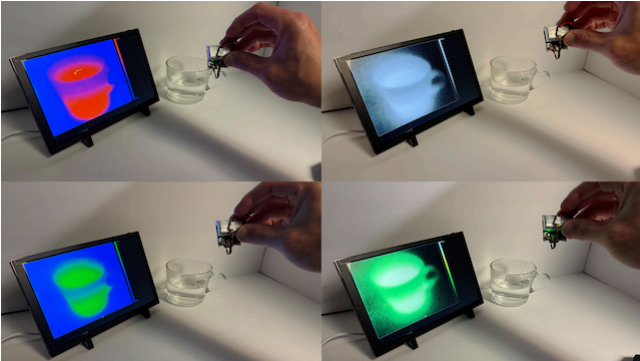
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🔥

ESP32 Thermal Camera Viewer with Node.js Web Interface

This project enables real-time thermal frame streaming from an ESP32-based thermal imaging camera (MI0802) to a web browser via TCP and WebSocket. It features a full-screen heatmap viewer, FPS tracking, temperature overlays, and charting.



ESP32 Thermal Camera powered by 3.7V Li-Po Battery

Project Structure

ESP32_Thermal_Camera/

├── senxorESP32S3/ # ESP32-S3 firmware (ESP-IDF)

├── Node_Thermal_TCP/ # Node.js TCP Client + Web UI

└── README.md

🚀

ESP32 Firmware (ESP-IDF)

The ESP32 source code has been modified based on the code provided by Waveshare. Please note that all rights to the code belong to Waveshare, which provides this code.

✓

Requirements

- Waveshare Thermal-Camera-ESP32-Module [\[https://www.waveshare.com/wiki/Thermal-Camera-ESP32-Module\]](https://www.waveshare.com/wiki/Thermal-Camera-ESP32-Module)
- ESP-IDF version 5.2.x only

🔧

Setup Instructions

1. Navigate to the ESP32 project folder:

cd senxorESP32S3

2. Configure the firmware:

idf.py menuconfig

- 🚨 **Important:**
Set your Wi-Fi SSID and password via `Example Configuration` menu.

3. Flash the firmware:

```
idf.py build
idf.py flash monitor
```

✔ Ensure your ESP32 connects to Wi-Fi and starts its TCP server on port `3333`.

🖥️ Node.js Web Viewer

✔ Requirements

- Node.js v16 or later
- A modern browser (Chrome, Edge, Firefox)

📦 Setup Instructions

1. Go to the Node.js project folder:

```
cd Node_Thermal_TCP
```

2. Install dependencies:

```
npm install
```

3. Open `client.js` and **edit the ESP32 IP address**:

```
const ESP32_HOST = "192.168.x.x"; // ← Your ESP32's local IP
```

⚠ Be sure to update this! Otherwise TCP connection will fail.

4. Start the client:

```
node client.js
```

5. Visit the web viewer in your browser:

```
http://localhost:8080
```

🌐 Features

- 🔥 Real-time 80x62 thermal frame rendering
- 🖼️ Fullscreen view with canvas scaling
- 🌈 Color map presets: Inferno, Plasma, Hot, Fire & Ice, etc.
- 📈 Live min/max temperature graph
- 🎯 Max temperature overlay + label
- 📏 Vertical range bar with color matching
- 📷 Screenshot saving
- ↔ Flip left-right toggle
- 🌡️ Celsius / Fahrenheit support

📄 License

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